

Boundary-Mode Topographic Cosmology: A Reduced-Response Bridge from Berger–Hopf Geometry to Late-Time Structure Growth

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Abstract

We develop a late-time cosmological bridge between a scalar topographic effective field theory on compact spatial sections $S^3(R_H)$ and the constraint-reduced response architecture of the Berger–Hopf program. The construction keeps a covariant cubic-Galileon background with second-order time evolution and introduces the higher-spatial-gradient response as the low-energy Schur/Feshbach reduction of a stable physical complement rather than as a fundamental higher-time-derivative interaction. On S^3 , scalar harmonics obey $-\nabla^2 Y_n = n(n+2)Y_n/R_H^2$. The reduced response can therefore soften the lowest nonconstant $n=1$ branch while stiffening $n \geq 2$, providing a mechanism for a large low-redshift dipolar susceptibility without a comparable quadrupole or large modification of ordinary redshift-space-distortion modes. We define a reference branch R1, specify an auditable preregistration protocol, and derive prospective growth and lensing targets. The low-redshift supernova amplitude is used only as an observational normalization of the $n=1$ response; the curvature forecast and structure-growth predictions are kept logically separate. The framework is explicitly falsifiable through spatial curvature, harmonic hierarchy, fixed-axis anisotropy, growth, and lensing tests.

1 Introduction

Late-time cosmological tensions and directional residuals motivate tests that are capable of separating local structure, survey geometry, and genuinely new large-scale dynamics. The scalar-topographic program introduced a compact hyperspherical parent $S^3(R_H)$ and a long-wavelength scalar response whose largest observational effect occurs at low redshift [1]. A later covariant formulation placed the topographic scalar in the cubic-Galileon sector of Horndeski theory, preserving second-order time evolution while allowing nontrivial late-time scalar–metric braiding.

The present work asks a narrower question: can the long-wavelength topographic response be represented as the large-scale branch of the same constraint-reduced response logic used in the Berger–Hopf program, without assuming that the full microscopic BHSM construction has already been compactified into cosmology? We answer this at the level of a controlled bridge model.

The central mathematical chain is

$$S[\Phi] \longrightarrow \hat{\mathcal{H}}_{\text{phys}} \longrightarrow \mathcal{K}_{\text{topo}} \longrightarrow \lambda_{\text{topo}} \longrightarrow T_n \longrightarrow \delta g_{\mu\nu} \longrightarrow \{D_L, H, f\sigma_8, \Phi + \Psi\}.$$

The distinction between established input and bridge extension is essential. The S^3 harmonic spectrum and cubic-Galileon background are treated as the cosmological starting point. The Schur/Feshbach reduction is an exact mathematical operation once a physical complement is

specified. The identification of its low-energy coefficients with the topographic response is a bridge assumption to be tested, not a declaration that the complete microscopic BSM action has already been reduced to four-dimensional cosmology.

The paper makes four main contributions. First, it standardizes the S^3 harmonic convention and shows how the observed sky dipole arises from the four-dimensional $n = 1$ eigenspace rather than by identifying a radial mode with an angular dipole. Second, it derives a reduced kernel in which a positive higher-spatial-gradient term can emerge after integrating out a stable complement. Third, it identifies a parameter interval in which the $n = 1$ response is selectively softened while all $n \geq 2$ harmonics are stiffened. Fourth, it gives an executable preregistration protocol for curvature, supernova anisotropy, growth, and lensing tests.

The claim boundary is therefore:

This work does not assume that the scalar-topographic cosmological field has already been derived from the complete microscopic Berger–Hopf action. Exact reductions are distinguished from cross-scale identifications and exploratory reference-branch assumptions. Prospective observables are frozen before comparison with designated independent data.

2 Hyperspherical Parent and Harmonic Convention

We use a closed FLRW geometry

$$ds^2 = -dt^2 + a^2(t)R_H^2 \left[d\chi^2 + \sin^2 \chi d\Omega_2^2 \right],$$

with $a_0 = 1$. Thus R_H is the present-day comoving curvature radius and the physical curvature radius is aR_H .

Scalar harmonics satisfy

$$-\nabla_\gamma^2 Y_n = \frac{n(n+2)}{R_H^2} Y_n, \quad n = 0, 1, 2, \dots,$$

with degeneracy $(n+1)^2$. The physical spatial eigenvalue entering the time-dependent perturbation equations is

$$p_n(a) = \frac{n(n+2)}{a^2 R_H^2}.$$

The first nonconstant eigenspace is therefore

$$p_1 = \frac{3}{a^2 R_H^2}.$$

This convention is used throughout. Older scalar-topographic notes employed more than one normalization for the lowest mode; no numerical forecast is carried forward here without first converting it to the convention above.

2.1 Exact light-cone form of the $n = 1$ mode

Embed the unit three-sphere in $\mathbb{R}^4$. The $n = 1$ harmonics are the four coordinate functions restricted to S^3 . A general real $n = 1$ mode can therefore be written

$$T_1(\chi, \hat{\mathbf{n}}) = A [\cos \alpha \cos \chi + \sin \alpha \sin \chi (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}})].$$

At fixed redshift this contains an exact sky monopole plus an exact sky dipole and no higher angular multipoles. The dipole part is

$$T_{1,\text{dip}}(z, \hat{\mathbf{n}}) = A \sin \alpha \sin \left[\frac{\chi(z)}{R_H} \right] (\hat{\mathbf{n}} \cdot \hat{\mathbf{p}}),$$

where $\chi(z)$ denotes the comoving radial distance in dimensional units when written inside the sine as χ/R_H .

This removes an ambiguity present in simplified radial descriptions: the zonal term $\cos \chi$ alone is isotropic for an observer at its pole. A sky dipole is supplied by a generic orientation of the full $n = 1$ eigenspace relative to the observer.

3 Constraint-Reduced Response and the Topographic Kernel

Let $\hat{\mathcal{H}}$ be the physical Hessian after constraints and gauge directions have been removed. For a selected topographic branch P and physical complement Q , the exact Schur/Feshbach reduced operator is

$$\mathcal{K} = P \left[\hat{\mathcal{H}} - \hat{\mathcal{H}}Q(Q\hat{\mathcal{H}}Q)^{-1}Q\hat{\mathcal{H}} \right] P.$$

For one scalar harmonic, write the quadratic block as

$$\hat{\mathcal{H}}_n = \begin{pmatrix} A(p) & C(p) \\ C^\dagger(p) & D(p) \end{pmatrix}.$$

The exact reduced stiffness is

$$K_{\text{red}}(p) = A(p) - C(p)D^{-1}(p)C^\dagger(p).$$

A minimal long-wavelength realization is

$$A(p) = Z_0 p, \quad D(p) = M_c^2 + c_c p, \quad C(p) = g\sqrt{p}.$$

Then

$$K_{\text{red}}(p) = Z_0 p - \frac{g^2 p}{M_c^2 + c_c p}.$$

For $c_c p \ll M_c^2$,

$$K_{\text{red}}(p) = \left(Z_0 - \frac{g^2}{M_c^2} \right) p + \frac{g^2 c_c}{M_c^4} p^2 + O(p^3).$$

Thus a positive p^2 stiffness can arise from a coupled second-order parent system after a stable complement is integrated out. In this interpretation the old static operator

$$-\nabla^2 + B\nabla^4$$

is a low-energy spatial derivative expansion, not evidence for a fundamental fourth-order time equation.

Define

$$\chi \equiv \frac{g^2}{M_c^2}, \quad L_c^2 \equiv \frac{c_c}{M_c^2}.$$

After canonical normalization of the leading spatial term, the reduced kernel may be written schematically as

$$K_{\text{topo}}(p) = m_T^2 + (c_s^2 - \chi)p + \chi L_c^2 p^2 + \dots$$

The signs of the temporal kinetic term and of the full physical gradient matrix must be checked independently; a positive spatial p^2 coefficient by itself is not a complete ghost analysis.

4 Cubic-Galileon Scalar–Metric Bridge

The cosmological parent action is taken to be

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R - \frac{1}{2} \nabla_\mu T \nabla^\mu T + \frac{1}{\Lambda^3} \square T (\nabla T)^2 - V(T) + \mathcal{L}_m \right].$$

Using

$$X \equiv -\frac{1}{2} \nabla_\mu T \nabla^\mu T,$$

this corresponds to

$$G_2 = X - V(T), \quad G_3 = \frac{2X}{\Lambda^3}, \quad G_4 = \frac{M_{\text{Pl}}^2}{2}, \quad G_5 = 0$$

in the convention $\mathcal{L}_3 = -G_3 \square T$.

The model belongs to the kinetic-gravity-braiding subclass of Horndeski theory [2]. Since G_4 is constant and $G_5 = 0$,

$$M_*^2 = M_{\text{Pl}}^2, \quad \alpha_M = 0, \quad \alpha_T = 0.$$

The scalar–metric mixing is controlled by the braiding function

$$\alpha_B = \frac{2\dot{T}^3}{H M_{\text{Pl}}^2 \Lambda^3},$$

and

$$\alpha_K = \frac{\dot{T}^2}{H^2 M_{\text{Pl}}^2} + 6\alpha_B.$$

The scalar no-ghost combination is

$$D_{\text{kin}} = \alpha_K + \frac{3}{2} \alpha_B^2.$$

For numerical work we softly break shift symmetry with

$$V(T) = V_0 - \mu^3 T.$$

The scalar energy density and pressure are

$$\rho_T = \frac{1}{2} \dot{T}^2 + V(T) + \frac{6H\dot{T}^3}{\Lambda^3},$$

$$P_T = \frac{1}{2} \dot{T}^2 - V(T) - \frac{2\dot{T}^2 \ddot{T}}{\Lambda^3}.$$

The shift current

$$J = \dot{T} + \frac{6H\dot{T}^2}{\Lambda^3}$$

obeys

$$\dot{J} + 3HJ = \mu^3.$$

The closed-FLRW background equation is

$$3M_{\text{Pl}}^2 \left(H^2 + \frac{1}{a^2 R_H^2} \right) = \rho_m + \rho_r + \rho_T.$$

The bridge does not require the bare Horndeski sound speed to become negative. Instead, the BHSM-inspired reduced response can lower the effective long-wavelength stiffness while the positive p^2 term stabilizes shorter wavelengths.

5 Selective Harmonic Response

Let the complement correlation length scale with the parent radius,

$$L_c(a) = aR_H \ell,$$

where ℓ is dimensionless. Since

$$p_n = \frac{n(n+2)}{a^2 R_H^2},$$

we have

$$L_c^2 p_n = \ell^2 n(n+2).$$

In the effectively massless topographic sector, define

$$K_n = p_n C_n$$

with

$$C_n(a) = c_s^2(a) - \chi(a) \left[1 - \ell^2 n(n+2) \right].$$

If

$$\frac{1}{8} < \ell^2 < \frac{1}{3},$$

then

$$1 - 3\ell^2 > 0$$

for $n = 1$, whereas

$$1 - \ell^2 n(n+2) < 0$$

for every $n \geq 2$. Hence increasing χ softens the $n = 1$ response but stiffens all higher harmonics:

$$C_1 < c_s^2, \quad C_n > c_s^2 \quad (n \geq 2).$$

This produces a stable near-critical alternative to a true gradient instability. If

$$0 < C_1 \ll 1,$$

then a sourced mode

$$T_1 \simeq \frac{S_1}{p_1 C_1}$$

can be strongly enhanced while all $n \geq 2$ responses remain regular.

For the quadrupole,

$$\frac{T_2}{T_1} = \frac{S_2}{S_1} \frac{3C_1}{8C_2}.$$

Therefore the response per unit source satisfies

$$\left| \frac{T_2/S_2}{T_1/S_1} \right| = \frac{3C_1}{8C_2} < \frac{3}{8}$$

on the stable selective-filter branch, with much stronger suppression as $C_1/C_2 \rightarrow 0$.

This is the primary harmonic null prediction of the model.

6 Observable Projection

The observable distance perturbation is not identified directly with the scalar amplitude. At linear order it is a light-cone projection of metric, velocity, and scalar perturbations:

$$\frac{\delta D_L}{\bar{D}_L} = \int d\chi \left[W_\Phi \Phi + W_\Psi \Psi + W_v v_\parallel + \dots \right].$$

For an $n = 1$ topographic source, the angular dependence factors into

$$\hat{\mathbf{n}} \cdot \hat{\mathbf{p}},$$

so that

$$\frac{\delta D_L}{\bar{D}_L} = \epsilon_D(z) \hat{\mathbf{n}} \cdot \hat{\mathbf{p}}.$$

For supernovae,

$$\mu = 5 \log_{10} D_L + \text{const},$$

and therefore a small magnitude dipole A_μ corresponds to

$$\frac{\delta D_L}{\bar{D}_L} = \frac{\ln 10}{5} A_\mu \cos \theta.$$

For structure growth we define

$$k^2 \Psi = -4\pi G a^2 \mu(k, a) \rho_m \delta_m, \quad \Phi = \gamma(k, a) \Psi,$$

and

$$\Sigma(k, a) = \frac{\mu(k, a)}{2} [1 + \gamma(k, a)].$$

The matter growth factor obeys

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \Omega_m(a) \mu(k, a) D = 0,$$

with prime denoting $d/d \ln a$. The observable is

$$f \sigma_8(k, z) = \frac{D'}{D} D \sigma_{8, \text{ini}} = D'(k, z) \sigma_{8, \text{ini}}.$$

The quasi-static limit is used only on scales inside the scalar sound horizon [3]. Horizon-scale $n = 1$ observables are treated through the full long-wavelength response rather than by extrapolating the quasi-static formulas.

7 Predictions and Falsification Criteria

7.1 Low-redshift calibration

The existing Pantheon+SH0ES tomographic analysis provides the observational normalization

$$A_\mu^\star = -0.0412 \pm 0.009$$

in the bin

$$0.01 < z < 0.03,$$

with best-fit direction

$$(\text{RA}, \text{Dec}) = (211.48^\circ, -12.81^\circ).$$

This corresponds to

$$\frac{\delta D_L}{D_L} = (-0.01895 \pm 0.00414) \cos \theta,$$

and at sufficiently low redshift,

$$\frac{\delta H}{H} = (+0.01895 \pm 0.00414) \cos \theta.$$

This datum normalizes only the observed $n = 1$ response,

$$A_\mu = \frac{\mathcal{K}_{\text{SN}} S_1}{p_1 C_1},$$

and does not independently determine S_1 , $\mathcal{K}_{\text{SN}}$, or C_1 .

7.2 Curvature

The pre-existing prospective curvature forecast is retained as

$$\boxed{\Omega_k = -0.018 \pm 0.004}.$$

At $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$,

$$R_H = \frac{c}{H_0 \sqrt{|\Omega_k|}} \simeq 33.15 \text{ Gpc}.$$

The sign prediction is $\Omega_k < 0$ in the usual FLRW curvature convention.

7.3 Harmonic hierarchy

The bridge predicts selective $n = 1$ softening with no analogous $n \geq 2$ enhancement. A future source-normalized $n = 2$ susceptibility comparable to or larger than the $n = 1$ susceptibility would reject the minimal harmonic-filter branch.

7.4 Fixed-axis test

A prospective independent supernova sample should first test the frozen axis

$$(211.48^\circ, -12.81^\circ)$$

rather than scanning freely over the sky. A free-axis search may be reported secondarily.

7.5 High- n recovery

At large n ,

$$C_n \simeq \chi \ell^2 n^2,$$

so the extra scalar susceptibility falls as n^{-2} . The model therefore predicts a strong separation between the horizon-scale dipole and conventional clustering modes.

7.6 Failure conditions

The minimal bridge is disfavored if one or more of the following are established with adequate precision:

1. independent curvature measurements exclude the frozen negative- Ω_k forecast;
2. the source-normalized $n = 2$ response is comparable to or exceeds $n = 1$;
3. an independent low-redshift sample excludes the frozen dipole axis while finding a stable incompatible physical axis;
4. a coherent dipole of comparable amplitude persists into a redshift regime in which the model predicts loss of enhanced susceptibility;
5. ordinary RSD modes show a large scale-independent modification incompatible with the high- n suppression;
6. lensing requires a metric response incompatible with the same scalar propagator that controls matter clustering.

8 Numerical Implementation and Preregistration Protocol

The numerical pipeline is separated into

$$\boxed{\text{background} \rightarrow \text{physical scalar response} \rightarrow \text{observables} \rightarrow \text{comparison}.}$$

Prediction generation and data comparison are implemented in separate scripts.

8.1 Background system

Let $v = \dot{T}$. Expanding the scalar current equation gives

$$\left(1 + \frac{12Hv}{\Lambda^3}\right) \dot{v} + \frac{6v^2}{\Lambda^3} \dot{H} = \mu^3 - 3H \left(v + \frac{6Hv^2}{\Lambda^3}\right).$$

The Raychaudhuri equation gives

$$\frac{2v^2}{\Lambda^3} \dot{v} - 2M_{\text{Pl}}^2 \dot{H} = \rho_m + \frac{4}{3}\rho_r + v^2 + \frac{6Hv^3}{\Lambda^3} - \frac{2M_{\text{Pl}}^2}{a^2 R_H^2}.$$

At each integration step these equations form a 2×2 linear system for $(\dot{v}, \dot{H})$. The remaining equations are

$$\dot{T} = v, \quad \dot{a} = aH.$$

The Friedmann equation is retained as an independent residual check:

$$3M_{\text{Pl}}^2 \left(H^2 + \frac{1}{a^2 R_H^2} \right) - (\rho_m + \rho_r + \rho_T) = 0.$$

8.2 Dimensionless variables

We integrate in $N = \ln a$ with

$$E = \frac{H}{H_0}, \quad \phi = \frac{T}{M_{\text{Pl}}}, \quad \lambda = \frac{\Lambda^3}{M_{\text{Pl}} H_0^2}, \quad q = \frac{\mu^3}{M_{\text{Pl}} H_0^2}.$$

The exploratory reference branch R1 uses

$$\Omega_{m0} = 0.31, \quad \Omega_{r0} = 9.0 \times 10^{-5}, \quad \Omega_{k0} = -0.018, \quad \lambda = 1, \quad q = 3.$$

The constant V_0 is determined only by the shooting condition $E(a = 1) = 1$.

8.3 Initial conditions

The numerical integration begins at

$$z_i = 100.$$

The decaying homogeneous current solution is removed using the matter-era particular solution

$$J_i \simeq \frac{2\mu^3}{9H_i}.$$

The initial scalar velocity is the positive small-velocity root of

$$v_i + \frac{6H_i v_i^2}{\Lambda^3} = J_i.$$

8.4 Growth grid

Before prospective comparison, predictions are generated on the fixed grid

$$k = \{0.02, 0.05, 0.10, 0.15, 0.20\} \text{ Mpc}^{-1}$$

and

$$z = \{0.5, 0.8, 1.0, 1.5, 2.1\}.$$

A deterministic JSON manifest containing the action version, parameters, numerical tolerances, calibration values, and predicted observables is hashed with SHA-256. Any later scientific or numerical revision creates a new manifest version rather than overwriting the previous freeze.

9 Results: Reference Branch R1

R1 is an exploratory reference branch rather than a uniquely derived BHSM cosmology. Its purpose is to demonstrate that the bridge can simultaneously realize late-time scalar activation, a selectively enhanced $n = 1$ response, and near-standard high- n structure growth.

For

$$\Omega_{m0} = 0.31, \quad \Omega_{r0} = 9.0 \times 10^{-5}, \quad \Omega_{k0} = -0.018, \quad \lambda = 1, \quad q = 3,$$

the shooting condition $E(1) = 1$ gives

$$\frac{V_0}{M_{\text{Pl}}^2 H_0^2} = 2.39861.$$

z	α_B	D_{kin}	Ω_T
10	2.07×10^{-7}	1.64×10^{-6}	0.00194
3	8.60×10^{-5}	6.80×10^{-4}	0.03915
2.1	3.72×10^{-4}	0.00294	0.08060
1.5	0.00122	0.00959	0.14321
1.0	0.00380	0.02988	0.24551
0.8	0.00622	0.04889	0.30787
0.5	0.01346	0.10536	0.43200
0.3	0.02271	0.17740	0.53538
0	0.04940	0.38485	0.70791

Table 1: Reference-branch background evolution.

The scalar braiding and no-ghost combination evolve as shown in Table 1.

The scalar equation of state evolves from nearly vacuum-like behavior at high redshift to

$$w_T(z=0) \simeq -0.906,$$

and the background acceleration transition occurs at

$$z_{\text{acc}} \simeq 0.683.$$

Relative to a curved Λ CDM model with identical $H_0, \Omega_{m0}, \Omega_{r0}, \Omega_{k0}$, the R1 expansion rate differs by only order one percent over the late-time interval. The maximum difference in the tabulated range is about 1.7% near $z \sim 0.5$.

9.1 Prospective growth target

In the minimal high- n realization, the direct topographic fifth-force term is negligible on ordinary RSD modes and the dominant growth shift is produced by the altered background history. With identical matter-era normalization,

$$R_{f\sigma_8}(z) = \frac{f\sigma_{8,\text{R1}}}{f\sigma_{8,\Lambda\text{CDM}}}$$

is

The central phenomenological result is therefore not a large generic modification of gravity. It is the combination

enhanced horizon-scale $n = 1$ susceptibility + rapid high- n recovery + few-percent late-time growth shift.

10 Reference-Branch Figures

11 Discussion

The bridge developed here differs from several common late-time alternatives in a specific way. A local-flow explanation can generate a low-redshift dipole but does not by itself predict a compact S^3 harmonic hierarchy, a frozen curvature sign, and a linked high- n response. Ordinary quintessence has a canonical smooth gradient sector and does not naturally single out one discrete

z	$R_{f\sigma_8}$	suppression
2.1	0.99346	0.65%
1.5	0.98869	1.13%
1.0	0.98164	1.84%
0.8	0.97795	2.20%
0.5	0.97271	2.73%
0.3	0.97168	2.83%
0.0	0.98482	1.52%

Table 2: R1 growth forecast relative to matched curved Λ CDM. These values are reference-branch predictions and become formally preregistered only when reproduced by the canonical code and written to the hashed manifest.

global spatial harmonic. A pure cubic Galileon supplies kinetic braiding but, without an additional scale-selective completion, a negative bare gradient coefficient would not isolate only the deep infrared. The reduced-response term introduced here instead permits the bare Horndeski branch to remain kinetically healthy while the physical $n = 1$ susceptibility is selectively enhanced.

The compact spectrum is central. With $R_H \simeq 33$ Gpc, ordinary RSD wavenumbers correspond to harmonics hundreds or thousands of levels above $n = 1$. This makes it possible for a large global dipolar response to coexist with nearly standard galaxy clustering. The model is therefore not equivalent to a scale-independent fifth force.

The same mechanism also clarifies the role of BHSM. The manuscript does not require that a cosmological scalar be identified directly with one microscopic Berger–Hopf field. What is imported is the reduced-response logic: a selected physical branch is evaluated only after the coupled complement has adjusted. In a derivative expansion, this produces a negative long-wave Schur correction together with a positive higher-spatial-gradient stiffness. That is the cross-scale structural claim made here.

Several limitations remain. The source law S_1 and the light-cone projection kernel must be derived in a complete perturbation treatment before the Pantheon normalization can be inverted into a unique microscopic susceptibility. The minimal no-slip closure used for some exploratory lensing estimates is not assumed to be a final theorem. The R1 values $q = 3$ and $\lambda = 1$ are reference-branch choices, not yet action-derived constants. The curvature forecast should be tested with directional-aware likelihoods and not retuned in response to a discrepant isotropic estimator.

These limitations are scientifically useful because each corresponds to a concrete next calculation rather than an unrestricted fitting freedom.

12 Conclusions

We have constructed a reduced-response bridge between hyperspherical scalar-topographic cosmology and the Berger–Hopf response architecture. The model retains a covariant cubic-Galileon background and interprets the higher-spatial-gradient topographic stiffness as the low-energy effect of integrating out a stable physical complement.

On $S^3(R_H)$, the exact scalar spectrum

$$p_n = \frac{n(n+2)}{a^2 R_H^2}$$

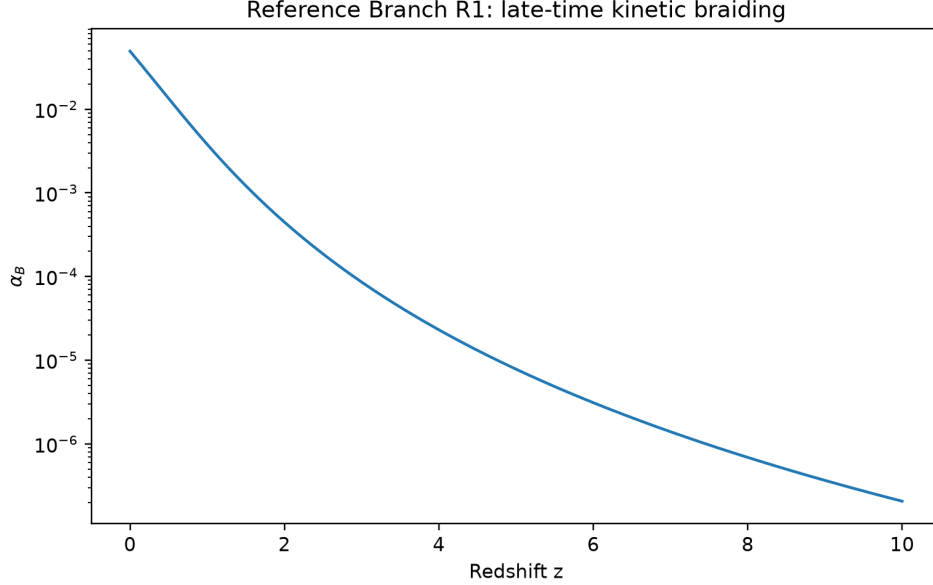


Figure 1: Kinetic-braiding function $\alpha_B(z)$ for Reference Branch R1. The logarithmic vertical scale emphasizes the suppression of the braiding sector toward high redshift.

allows a simple harmonic filter. For

$$\frac{1}{8} < \ell^2 < \frac{1}{3},$$

the same response that softens $n = 1$ stiffens every $n \geq 2$ mode. This yields a large dipolar susceptibility without requiring a comparable quadrupole and without forcing a large modification of ordinary structure-growth modes.

The low-redshift Pantheon+SH0ES dipole is used only to normalize the observed $n = 1$ response. The pre-existing curvature forecast

$$\Omega_k = -0.018 \pm 0.004$$

is kept separate. An exploratory reference branch R1 gives a late-time braiding component that is negligible at high redshift and predicts only a few-percent suppression of $f\sigma_8$ relative to matched curved Λ CDM over the main survey range.

The strongest prediction is therefore a pattern rather than a single number:

$$\boxed{\text{strong } n = 1 \text{ response} \oplus \text{suppressed } n \geq 2 \text{ response} \oplus \text{near-GR high-}k \text{ growth.}}$$

The accompanying preregistration protocol is designed so that curvature, harmonic hierarchy, fixed-axis anisotropy, growth, and lensing can falsify the bridge without post-comparison retuning.

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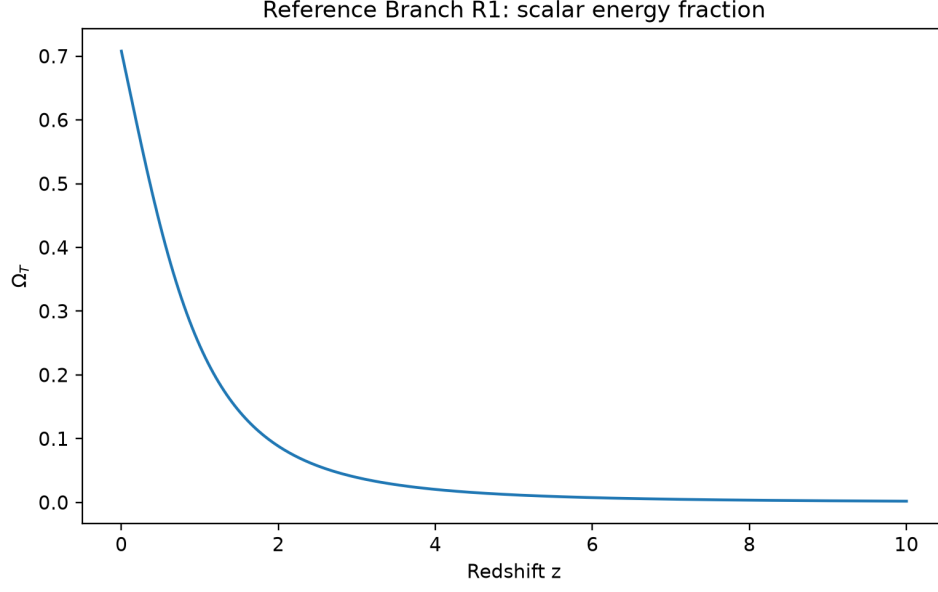


Figure 2: Scalar energy fraction $\Omega_T(z)$ for R1. The scalar component is subdominant at high redshift and becomes dynamically important only at late times.

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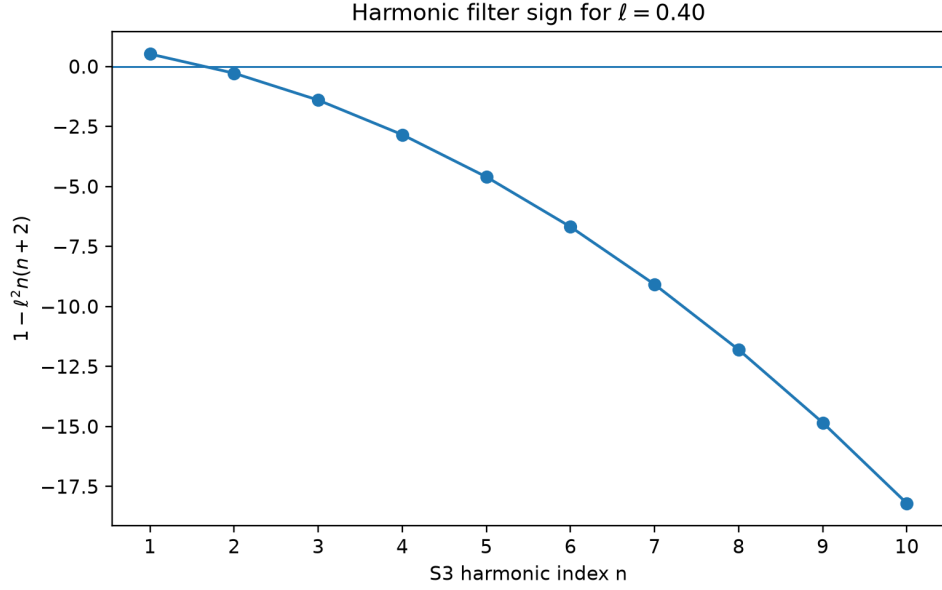


Figure 3: Sign of the Schur-response factor $1 - \ell^2 n(n+2)$ for the illustrative filtering value $\ell = 0.40$. The $n = 1$ branch lies on the softening side while every displayed $n \geq 2$ mode lies on the stiffening side.

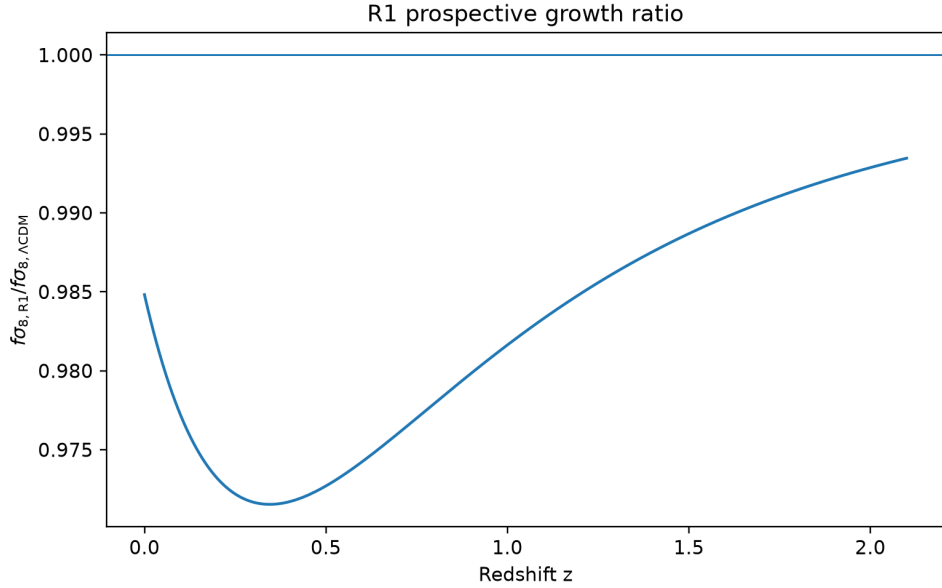


Figure 4: Prospective R1 growth ratio relative to a matched curved- Λ CDM background with the same early normalization.